TELECOM CUSTOMER CHURN ANALYSIS REPORT

1. Project Overview

This project analyzes customer behavior using a telecom customer churn dataset. The objective is to identify factors influencing customer churn, understand usage patterns, and provide business insights to improve customer retention. The project combines Python, PostgreSQL, and Power BI.

2. Dataset Summary

- **Rows**: 3,333
- **Columns**: 20

Key Features:
- Customer Information
- Service Plans
- Telecommunication Usage
- Customer Support
- Target Variable (Churn)

Missing Values: None.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in python:

- **Data loading:** Imported the dataset using **pandas.**
- **Initial Exploration**: Used **df.info ()** to check structure and. **describe ()** for summary statistics

[4]:

	State	Account length	Area code	International plan	Voice mail plan	Number vmail messages	Total day minutes	Total day calls	Total day charge	Total eve minutes	Total eve calls
0	KS	128	415	No	Yes	25	265.1	110	45.07	197.4	99
1	OH	107	415	No	Yes	26	161.6	123	27.47	195.5	103
2	NJ	137	415	No	No	0	243.4	114	41.38	121.2	110
3	OH	84	408	Yes	No	0	299.4	71	50.90	61.9	88
4	OK	75	415	Yes	No	0	166.7	113	28.34	148.3	122

Total eve calls	Total eve charge	Total night minutes	Total night calls	Total night charge	Total intl minutes	Total intl calls	Total intl charge	Customer service calls	Churn
99	16.78	244.7	91	11.01	10.0	3	2.70	1	False
103	16.62	254.4	103	11.45	13.7	3	3.70	1	False
110	10.30	162.6	104	7.32	12.2	5	3.29	0	False
88	5.26	196.9	89	8.86	6.6	7	1.78	2	False
122	12.61	186.9	121	8.41	10.1	3	2.73	3	False

Database Integration: Connected Python script to PostgreSQL and loaded the cleaned Data Frame into the database for SQL analysis.

4. Business-Oriented Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions:

Business Question 1

Which telecommunication services are used the most by customers?

SQL Showing rows: 1 to 1

	avg_day_minutes numeric	avg_evening_minutes numeric	avg_night_minutes numeric	avg_international_minutes numeric
1	179.78	200.98	200.87	10.24

Business Question 2

How many customers churned and how many remained?

Data Output Messages Notifications

	churn boolean	total_customers bigint	percentage numeric
1	false	2850	85.51
2	true	483	14.49

Business Question 3
What is the customer distribution by state?

	state text	total_customers bigint
1	WV	106
2	MN	84
3	NY	83
4	AL	80
5	OR	78
6	WI	78
7	OH	78
8	VA	77
9	WY	77
10	CT	74
11	ID	73
12	VT	73
13	MI	73
14	UT	72
15	TX	72
16	IN	71

Business Question 4
What is the average number of customer service calls?

	average_customer_service_calls numeric
1	1.56

Business Question 5
Which service plan is the most popular?

	state text	international_plan bigint	voice_mail_plan bigint
1	AK	4	16
2	AL	8	21
3	AR	8	14
4	AZ	3	19
5	CA	4	11
6	CO	4	19
7	CT	8	21
8	DC	5	18
9	DE	10	15
10	FL	8	20
11	GA	4	19
12	HI	3	15
13	IA	0	12
14	ID	6	28
15	IL	15	15

Business Question 6

What is the average revenue generated per customer?

	average_revenue numeric
1	59.45

Business Question 7

Do more customers have an International Plan or a Voice Mail Plan?

	service_plan text	total_customers bigint
1	International Pl...	323
2	Voice Mail Plan	922

Business Question 8

Which states have the highest number of non-churn customers?

	state text	retained_customers bigint
1	WV	96
2	VA	72
3	AL	72
4	WI	71
5	MN	69
6	OH	68
7	WY	68
8	NY	68
9	OR	67
10	VT	65
11	ID	64
12	IN	62
13	UT	62
14	CT	62
15	AZ	60

Business Question 9

Which account length range has the highest number of customers?

	account_length_group text	total_customers bigint
1	50-100 days	1338
2	101-150 days	1313
3	More than 150 days	356
4	Less than 50 days	326

5. Interactive Dashboard using Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.

6 Business Recommendations

- Most customers are retained.
- Evening calls are the most used.
- International calls are the least used.
- Voice Mail Plan is more popular.
- Average revenue per customer is approximately 60.

7. Business Recommendations

- Improve retention.
- Enhance customer support.

- Promote underused services.
- Focus on high-risk states.

8. Technologies Used

Python, Pandas, PostgreSQL, Power BI, GitHub

9. Conclusion

This project demonstrates an end-to-end data analytics workflow from Python to SQL and Power BI.